

$$X = \{2, 4, 6, 8, 10\}$$

$$\bar{X} = 6 \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$\frac{16 + 16 + 4 + 4}{4} = 10$$

$$\bar{X} \pm t_{0.05, r} \cdot SE$$

$$t = \frac{X - \mu}{s/\sqrt{n}}$$

$$n = 100$$

$$\bar{x} =$$

$$\sigma = 2$$

$$z = 1.96$$

$$\frac{2}{10} = 0.2 = SE$$

$$1.96 \cdot 0.2 = 0.392$$

1. A sample dataset of $n = 20$ observations is collected:

$\{12, 15, 18, 14, 16, 20, 22, 19, 17, 15, 13, 16, 18, 21, 23, 14, 16, 17, 19, 20\}$.

Calculate the sample mean ($\bar{x}$) and the sample variance (s^2). Then, test if the population mean is significantly greater than 15 using a one-sample t-test at $\alpha = 0.05$. What is the calculated t-statistic?

A. 2.54

B. 3.17

C. 4.12

D. 1.73

27
45
85
136
168
187
236
273
306
345

$$345/20 = 17,25$$

$$\text{mean} = 17,25$$

$$t_{0,025, 19} = 2,093 \cdot SE \cdot \frac{s}{\sqrt{20}}$$

$$1,96 \cdot 3/\sqrt{20}$$

$$\frac{15 - 17,25}{s/\sqrt{20}}$$

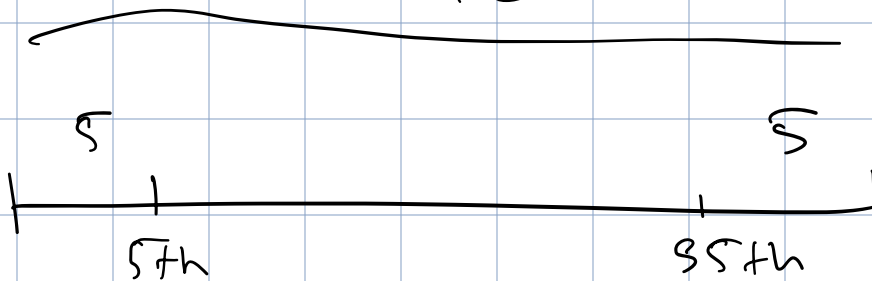
$$s = 3,02$$

$$1.96 = \frac{x - \mu}{\sigma}$$

$$1.96 \cdot 3$$

$$\frac{15 - 17.35}{3.22/\sqrt{20}}$$

100

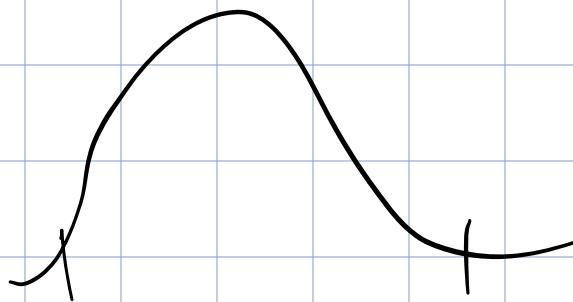


$$58.15 \pm SE. t$$

901

0.0048

0.9951



$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \quad Y = \begin{bmatrix} 2 \\ 5 \\ 5 \end{bmatrix}$$

$$Y = \beta_0 + \beta_1 X$$

$$\beta = (X^T X)^{-1} X^T Y$$

$$\begin{matrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} & \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} & = & \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} & = & X^T X = A \\ 2 \times 3 & 3 \times 2 & & 2 \times 2 & & \end{matrix}$$

$$\det(A) = 3 \cdot 14 - 6 \cdot 6 = 42 - 36 = 6$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} = \begin{bmatrix} 7/3 & -1 \\ -1 & 0.5 \end{bmatrix}$$

$$\begin{matrix} \begin{bmatrix} 7/3 & -1 \\ -1 & 0.5 \end{bmatrix} & \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} & = & \begin{bmatrix} 4/3 & 1/3 & -2/3 \\ -0.5 & 0 & 0.5 \end{bmatrix} \\ 2 \times 2 & 2 \times 3 & & 2 \times 3 \end{matrix}$$

$$\begin{bmatrix} 4/3 & 1/3 & -2/3 \\ -0.5 & 0 & 0.5 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 1.5 \end{bmatrix}$$

$2 \times 3 \qquad 3 \times 1 \qquad 2 \times 1$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

$$X = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 2 & 1 \end{bmatrix} \quad Y = \begin{bmatrix} 2 \\ 3 \\ 5 \\ 6 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$A = X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 6 & 2 \\ 6 & 10 & 3 \\ 2 & 3 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A)$$

$$\begin{aligned} & 80 + 36 + 36 \\ & - 40 - 36 - 36 \\ & = 4 = \det(A) \end{aligned}$$

$$\text{adj}(A)^T = \begin{bmatrix} 11 & -6 & -2 \\ -6 & 4 & 0 \\ -2 & 0 & 4 \end{bmatrix} = \text{adj}(A)$$

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 11 & -6 & -2 \\ -6 & 4 & 0 \\ -2 & 0 & 4 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 16 \\ 27 \\ 9 \end{bmatrix}$$

$$\beta = \frac{1}{4} \begin{bmatrix} 11 & -6 & -2 \\ -6 & 4 & 0 \\ -2 & 0 & 4 \end{bmatrix} \begin{bmatrix} 16 \\ 27 \\ 9 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -4 \\ 12 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix} \quad Y = \begin{bmatrix} 2 \\ 1 \\ 5 \\ 6 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 \end{bmatrix}$$

$$A = X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 4 & 4 \\ 4 & 6 & 4 \\ 4 & 4 & 6 \end{bmatrix}$$

3×4 4×3 3×3

$$A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A)^T$$

$$\begin{aligned} \det(A) &= 4 \cdot 6 \cdot 6 + 4 \cdot \cancel{4} \cdot 4 + 4 \cdot 4 \cdot \cancel{4} - 4 \cdot 6 \cdot 4 - 4 \cdot \cancel{4} \cdot 4 - 4 \cdot 4 \cdot 6 \\ &= 4(36 + 16 - 24 - 24) = 4 \cdot 4 = 16 \end{aligned}$$

$$\text{adj}(A)^T = \begin{bmatrix} 20 & -8 & -8 \\ -8 & 8 & 0 \\ -8 & 0 & 8 \end{bmatrix} = \text{adj}(A)$$

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 5 & -2 & -2 \\ -2 & 2 & 0 \\ -2 & 0 & 2 \end{bmatrix}$$

$$X^T Y = \begin{matrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix} & \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} & \hat{=} & \begin{bmatrix} 12 \\ 14 \\ 16 \end{bmatrix} \\ 3 \times 3 & 3 \times 1 & & 3 \times 1 \end{matrix}$$

$$B = \frac{1}{2} \begin{bmatrix} 5 & -2 & -2 \\ -2 & 2 & 0 \\ -2 & 0 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \\ 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \quad Y = \begin{bmatrix} 3 \\ 2 \\ 7 \end{bmatrix}$$

$$A = X^T X = \begin{matrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} & \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} & = & \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \\ 2 \times 3 & 3 \times 2 & & 2 \times 2 \end{matrix}$$

$$A^{-1} = \frac{1}{\det A} \cdot \text{adj}(A) = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}$$

$$X^T Y = \begin{matrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} & \begin{bmatrix} 3 \\ 2 \\ 7 \end{bmatrix} & \hat{=} & \begin{bmatrix} 12 \\ 28 \end{bmatrix} \\ 2 \times 3 & 3 \times 1 & & 2 \times 1 \end{matrix}$$

$$\frac{1}{3} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} 6 \\ 14 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \hat{\beta}$$

$$y = 2x_1 + \epsilon \quad X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \quad Y = \begin{bmatrix} 3 \\ 2 \\ 7 \end{bmatrix}$$

$$\epsilon = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$\epsilon^2 = \epsilon^T \epsilon = 6$$

$$s^2 = \frac{\epsilon^T \epsilon}{n-p} = \frac{6}{3-2} = 6$$

$$\text{var}(\hat{\beta}) = s^2 (X^T X)^{-1} = 6 \cdot \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}$$

$$\text{var} \beta_0 = 14$$

$$\text{var} \beta_1 = 3$$

$$X = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad Y = \begin{bmatrix} 2 \\ 3 \\ 2 \\ 1 \end{bmatrix}$$

$$A = X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det A} \cdot \text{adj}(A)^T = \frac{1}{4} \begin{bmatrix} 3 & -2 & -2 \\ -2 & 4 & 0 \\ -2 & 0 & 4 \end{bmatrix}$$

$$\det(A) = 6 + 4 + 4 - 8 - 4 - 8 = 4$$

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ 5 \\ 5 \end{bmatrix}$$

3×4 4×1 3×1

$$\beta = \frac{1}{4} \begin{bmatrix} 3 & -2 & -2 \\ -2 & 4 & 0 \\ -2 & 0 & 4 \end{bmatrix} \begin{bmatrix} 8 \\ 5 \\ 5 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

3×3 3×1 3×1

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

$$\epsilon = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \sum \epsilon^2 = 0,5$$

$$s^2 = \frac{0,5}{4-3} = 0,5$$

$$\begin{aligned} \text{var}(\beta) &= s^2 \cdot (X^T X)^{-1} = 0,5 \cdot \frac{1}{4} \cdot \begin{bmatrix} 3 & -2 & -2 \\ -2 & 4 & 0 \\ -2 & 0 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 3/8 & -1/4 & -1/4 \\ -1/4 & 1/2 & 0 \\ -1/4 & 0 & 1/2 \end{bmatrix} \end{aligned}$$

$$\text{var}(\beta_0) = 3/8$$

$$\text{var}(\beta_1) = 1/2$$

$$\text{var}(\beta_2) = 1/2$$

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$y = \begin{bmatrix} 3 \\ 5 \\ 2 \\ 1 \end{bmatrix}$$

$$A = X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A) =$$

$$\det(A) = 16$$

$$X = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$y = \begin{bmatrix} 3 \\ 5 \\ 2 \\ 1 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{4} \begin{bmatrix} 3 & -2 & -2 \\ -2 & 4 & 0 \\ -2 & 0 & 4 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 11 \\ 8 \\ 7 \end{bmatrix}$$

$$\frac{1}{4} \begin{bmatrix} 3 & -2 & -2 \\ -2 & 4 & 0 \\ -2 & 0 & 4 \end{bmatrix} \begin{bmatrix} 11 \\ 8 \\ 7 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 3 \\ 10 \\ 6 \end{bmatrix} = \begin{bmatrix} 3/4 \\ 5/2 \\ 3/2 \end{bmatrix} = \beta$$

$$e = \begin{bmatrix} -1/4 \\ -1/4 \\ 1/4 \end{bmatrix}$$

$$\sum e^2 = e^T e = 1/4$$

$$s^2 = \frac{1/4}{1} = 1/4$$

$$\text{var}(\beta) = s^2 A^{-1} = \frac{1}{4} \cdot \frac{1}{4} \cdot \begin{bmatrix} 3 & 4 & 4 \\ & 4 & 4 \\ & & 4 \end{bmatrix}$$

$$\text{var}(\beta_0) = \frac{3}{16} \quad \text{SE}(\beta_0) = \frac{\sqrt{3}}{4}$$

$$\text{var}(\beta_1) = \frac{1}{4} \quad \text{SE}(\beta_1) = \frac{1}{2}$$

$$\text{var}(\beta_2) = \frac{1}{4} \quad \text{SE}(\beta_2) = \frac{1}{2}$$